Remember

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j} = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT\ln x_i + RT\ln\gamma_i$$

Depression of freezing point

$$\ln \gamma_1 x_1 = \frac{-\Delta \underline{H}_1^{fus}}{RT_f^{\,2}}\Delta T$$

Approximate result for small ΔT

The Gibbs Phase Rule (revision)

We are interested in determining the equilibrium position of multi-component, multi- phase systems

e.g. what is the composition and pressure of a vapour in equilibrium with a binary liquid mixture of specified composition at a given temperature

Note we need to specify only two things (liquid composition and temperature) to completely define this system. The Gibbs phase rule tells us why this is the case

The Gibbs phase rule is concerned with the number of **intensive** properties required to specify a multicomponent, multiphase system **at equilibrium**.

Intensive and extensive properties come in conjugate pairs: a driving force (the intensive property) and the thing which flows (the extensive property) -

$P + V \qquad T + S \qquad \mu_i + n_i$

For a pure substance in a single phase, two **intensive** properties are necessary to define the state of the substance (e.g. P and T)

(Note that we also need to know how much of the substance we have - if we want to completely define the **system**.)

For a single phase system with C components, we require a further C–1 mole fractions

If we now consider P separate phases (each with C components), then we would need to know $P(C–1+2)$ intensive properties, assuming that there are no constraints at all

However, equilibrium imposes constraints

Thermal and mechanical equilibrium between each phase impose $2(P-1)$ constraints:

$$T^{I} = T^{II}$$ ← phase

$$T^{II} = T^{III}$$

……etc.

$$P^{I} = P^{II}$$

$$P^{II} = P^{III}$$

} not always true – see later

……etc.

Equilibrium with respect to mass transfer between the phases for each component imposes $C(P-1)$ additional constraints:

$$\mu_1^{I} = \mu_1^{II};\ \mu_2^{I} = \mu_2^{II};\ldots\ldots\ldots\mu_C^{I} = \mu_C^{II}$$

$$\mu_1^{II} = \mu_1^{III};\ \mu_2^{II} = \mu_2^{III};\ldots\ldots\ldots\mu_C^{II} = \mu_C^{III}$$ ← component

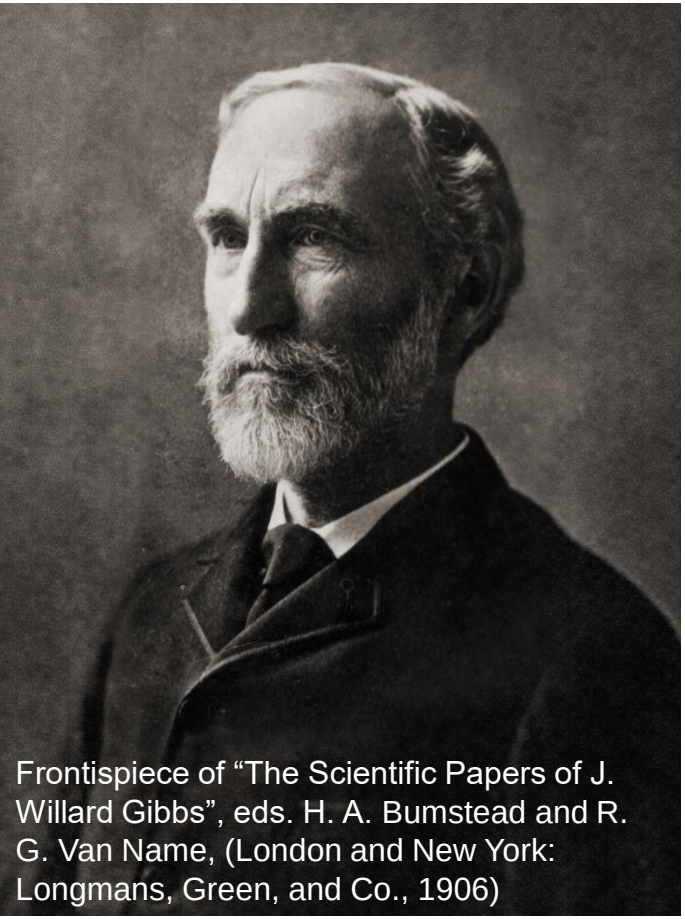
…..etc.

$$\mu_1^{P-1} = \mu_1^{P};\ \mu_2^{P-1} = \mu_2^{P};\ldots\ldots\ldots\mu_C^{P-1} = \mu_C^{P}$$

The total number of intensive variables required is thus:

$$F = P(C-1+2)-(P-1)(2+C)$$

i.e. $\boxed{F = C - P + 2}$ the Gibbs phase rule



Frontispiece of “The Scientific Papers of J. Willard Gibbs”, eds. H. A. Bumstead and R. G. Van Name, (London and New York: Longmans, Green, and Co., 1906)

“The greatest mind in American history.” Albert Einstein

Josiah Willard Gibbs,1839-1903

Professor of Mathematical Physics, Yale

“Father of Thermodynamics”

"A mathematician may say anything he pleases, but a physicist must be at least partially sane"



U.S postage stamps, American Scientists, first series, 2005 (bardostamps.com)

Examples

Pure liquid water + pure steam (n.b. no air present)

$C = 1 \quad P = 2$

$F = 1 - 2 + 2 = 1$

Only one degree of freedom define P, know T – unique boiling point

at given T, a line on the phase diagram

Pure liquid water + pure steam + pure ice

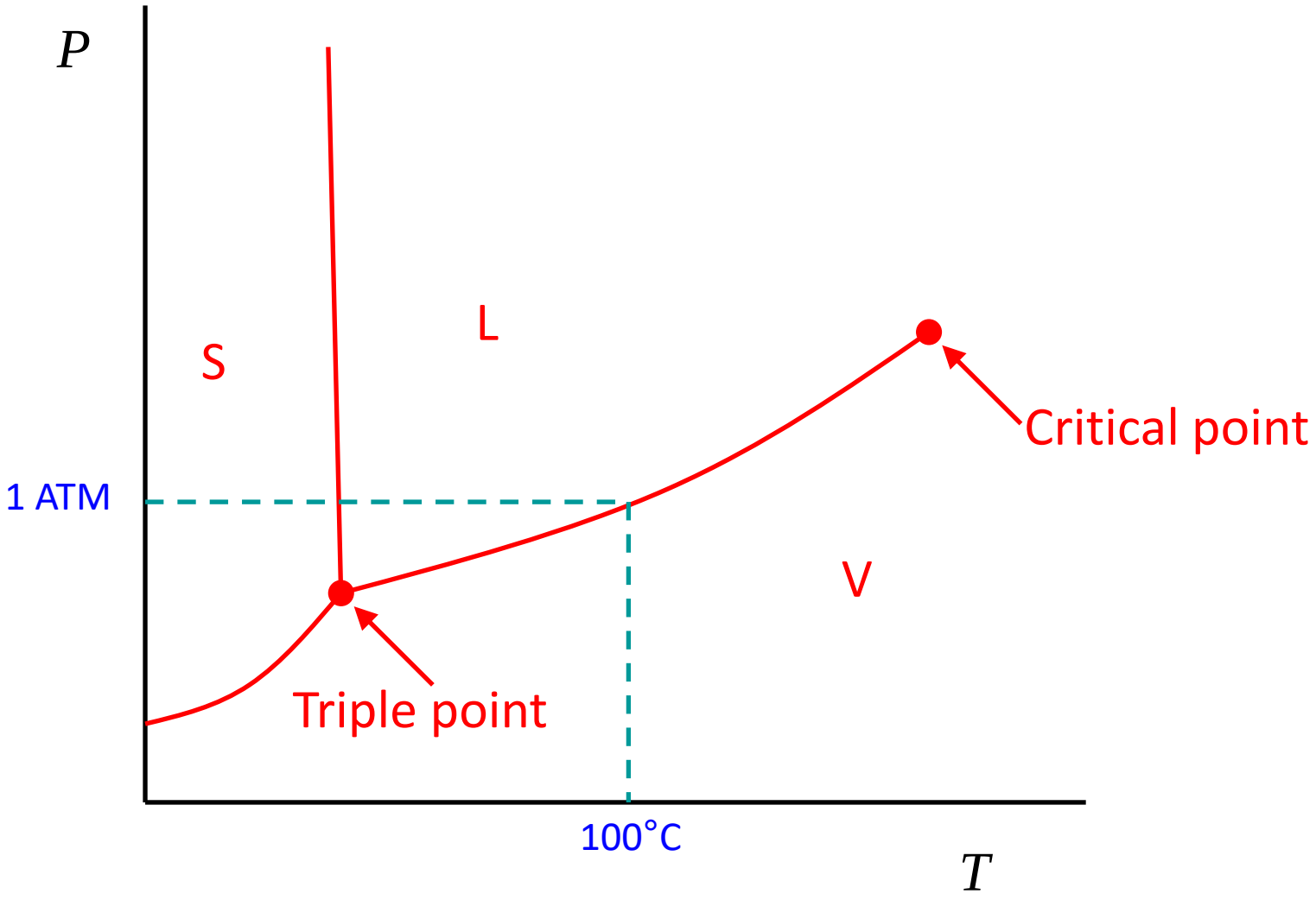
$C = 1 \quad P = 3$

$F = 1 - 3 + 2 = 0$

Triple point at fixed T, P (273.16K, 0.006 bar)

a point on the phase diagram

Pure H_2O phase diagram (schematic)



A couple of “Exceptions” to the Gibbs phase rule

Not really exceptions, but not all the equilibrium constraints apply

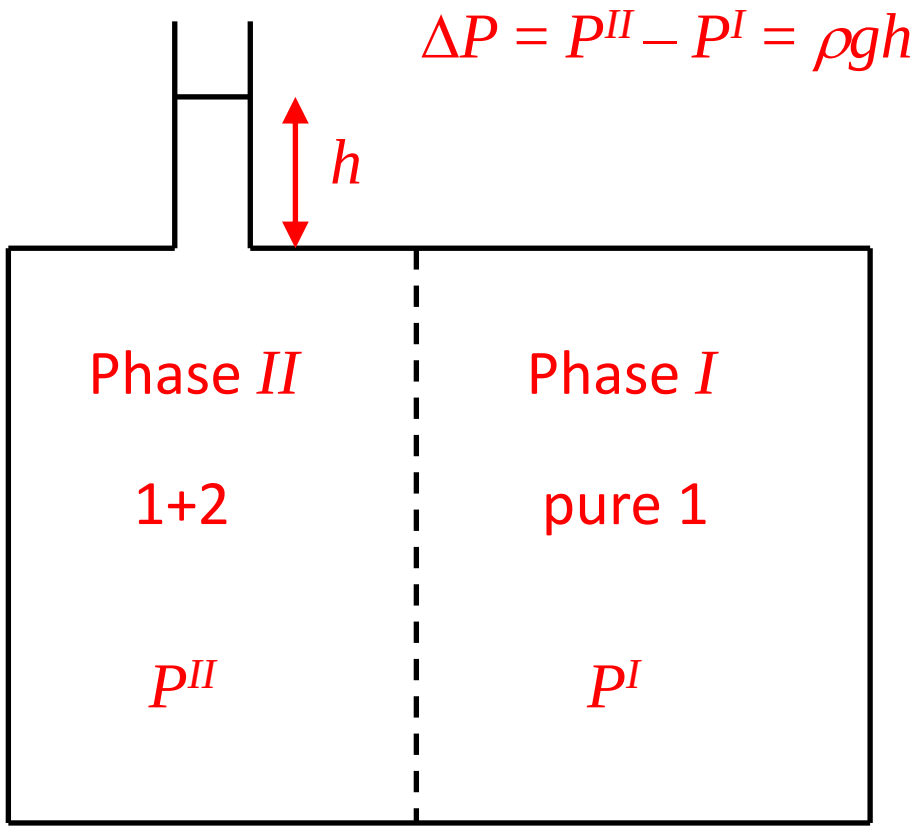
Osmotic equilibrium

Two liquids separated by a semi-permeable membrane



Binary mixture
component 1 can cross the membrane
component 2 cannot

Pure 1 on one side of the membrane



The Gibbs phase rule suggests that:

$$F = C - P + 2 = 2 - 2 + 2 = 2$$

However this is the wrong answer. Why?

It is not necessary for the chemical potential of component 2 to be the same on both sides of the membrane

We do not require the pressure to be the same on both sides

So in fact $F = 4$

We could freely specify the temperature, the pressure on one side of the membrane and the compositions on both sides of the membrane

Consider the chemical equilibrium of component 1 (which can move freely across the membrane)

We require:

$$\mu_1^I = \mu_1^{II}$$

phase

component

Phase I is pure component 1,

phase II also contains component 2

$$\mu_1^I = \underline{G}_1 = \underline{G}_1^0 + \int_{P^0}^{P^I} \underline{V}_1 dP \approx \underline{G}_1^0 + \underline{V}_1\left(P^I - P^0\right)$$

molar volume of pure 1

pure liquid 1 at T, P^0

pressure in phase I

liquid $\Rightarrow \underline{V}_1 \neq \text{fn}(P)$

and

$$\mu_1^{II} = \underline{G}_1 + RT \ln \gamma_1 x_1$$

non-ideal mixing

pure liquid 1 at T, P^{II}

ideal mixing

Setting $P^0 = P^{II}$

$$-RT \ln \gamma_1 x_1 = \underline{V}_1 \Delta P$$

where $\Delta P = P^{II} - P^{I}$ the osmotic pressure

Note that $x_1 < 1$ so the pressure is higher on the side containing impurities (i.e. ΔP is positive)

As for depression of freezing point, the osmotic pressure does not depend on the nature of component 2 (except via γ_1)

It is another colligative property

2 possibilities:

- use an activity coefficient model to get an accurate ΔP
- use ΔP to calculate γ_1 as fn(x_1)

If 2 is dilute,

then $\gamma_1 \approx 1$

and $\ln x_1 = \ln(1 - x_2) \approx -x_2$

Hence $\Delta P \approx \dfrac{RTx_2}{\underline{V}_1}$

c.f. ideal gas law $P = \dfrac{RT}{\underline{V}}$

c.f. depression of freezing point (dilute case) $\Delta T \approx \dfrac{RT_f{}^2 x_2}{\Delta \underline{H}_1^{fus}}$